Prompt Engineering Challenge

1. Summarising Notes

Prompt: Summarize the following notes into simple bullet points suitable for a BTech student. Highlight key concepts and important definitions.

Full AI Response:
• Artificial Intelligence (AI) is the simulation of human intelligence by machines.
• AI helps computers learn from data and improve performance.
• AI can reason, make decisions, and solve problems.
• AI is used in chatbots, recommendation systems, healthcare, and self-driving cars.
• AI aims to make machines perform tasks that normally require human intelligence.

Why the Prompt Worked: The prompt clearly defined the task, audience, and format.

2. Generating a Study Plan

Prompt: Create a 7-day study plan for a BTech student preparing for DBMS exams.

Full AI Response:
Day 1: Introduction to DBMS
Day 2: ER Model
Day 3: SQL Queries
Day 4: Normalization
Day 5: Transactions
Day 6: Indexing and Recovery
Day 7: Revision and Mock Test

Why the Prompt Worked: The prompt specified duration, subject, and structure.

3. Creating Interview Questions

Prompt: Generate 10 Java interview questions with answers for a fresher software developer.

Full AI Response:
1. What is Java?
2. What is JVM?
3. What is JDK?
4. What is JRE?
5. What is Inheritance?
6. What is Polymorphism?
7. What is Encapsulation?
8. What is Abstraction?
9. What is an Interface?
10. What is Exception Handling?

Why the Prompt Worked: The prompt clearly stated the technology and audience.

4. Writing a Resume Summary

Prompt: Write a professional resume summary for a final-year BTech student with skills in Java, Python, Salesforce, and Web Development.

Full AI Response:
Motivated and detail-oriented final-year BTech student with knowledge of Java, Python, Salesforce, and Web Development.

Why the Prompt Worked: The prompt provided educational background and skills.

5. Creating a Lesson Plan

Prompt: Create a one-hour lesson plan to teach Artificial Intelligence basics to beginners.

Full AI Response:
0–10 Minutes: Introduction to AI
10–20 Minutes: History of AI
20–35 Minutes: Applications
35–50 Minutes: Machine Learning Basics
50–60 Minutes: Discussion

Why the Prompt Worked: The prompt clearly defined topic, audience, and duration.